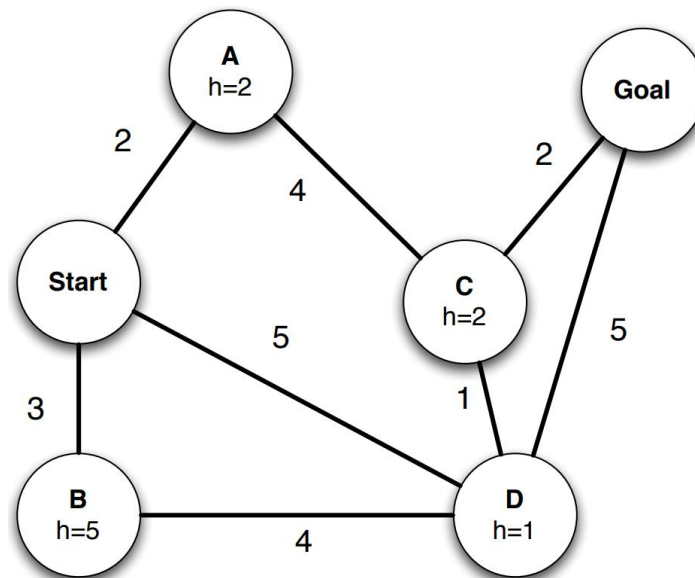


informed and uninformed search

1) (Graph search algorithms)

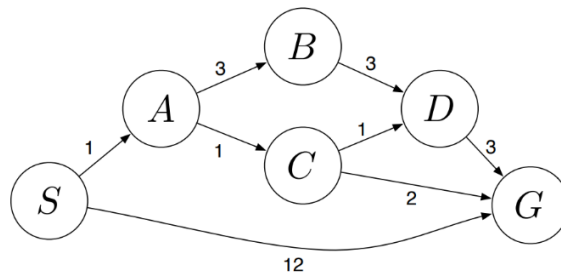
For each of the following graph search strategies, work out the order in which states are expanded, as well as the path returned by graph search. In all cases, assume ties resolve in such a way that states with earlier alphabetical order are expanded first. The start and goal state are S and G, respectively. Remember that in graph search, a state is expanded only once.



- a) Depth-first search.
- b) Breadth-first search.
- c) Uniform cost search.
- d) Greedy search with the heuristic h shown on the graph.
- e) A* search with the same heuristic.

2) (Evaluation of heuristics)

Answer the following questions about the search problem shown above. Break any ties alphabetically. For the questions that ask for a path, please give your answers in the form 'S – A – D – G.'



- What path would breadth-first graph search return for this search problem?
- What path would uniform cost graph search return for this search problem?
- What path would depth-first graph search return for this search problem?
- What path would A* graph search, using a consistent heuristic, return for this search problem?
- Consider the heuristics for this problem shown in the table below.

State	h_1	h_2
<i>S</i>	5	4
<i>A</i>	3	2
<i>B</i>	6	6
<i>C</i>	2	1
<i>D</i>	3	3
<i>G</i>	0	0

- Is h_1 admissible? Yes No
- Is h_1 consistent? Yes No
- Is h_2 admissible? Yes No
- Is h_2 consistent? Yes No

3) (Status space search)

Pacman and Ms. Pacman are lost in an $N \times N$ maze and would like to meet; they don't care where. In each time step, both simultaneously move in one of the following directions: {NORTH, SOUTH, EAST, WEST, STOP}. They do not alternate turns. You must devise a plan which positions them together, somewhere, in as few time steps as possible. Passing each other does not count as meeting; they must occupy the same square at the same time.

- a. Formally state this problem as a single-agent state-space search problem.

States:

Goal test:

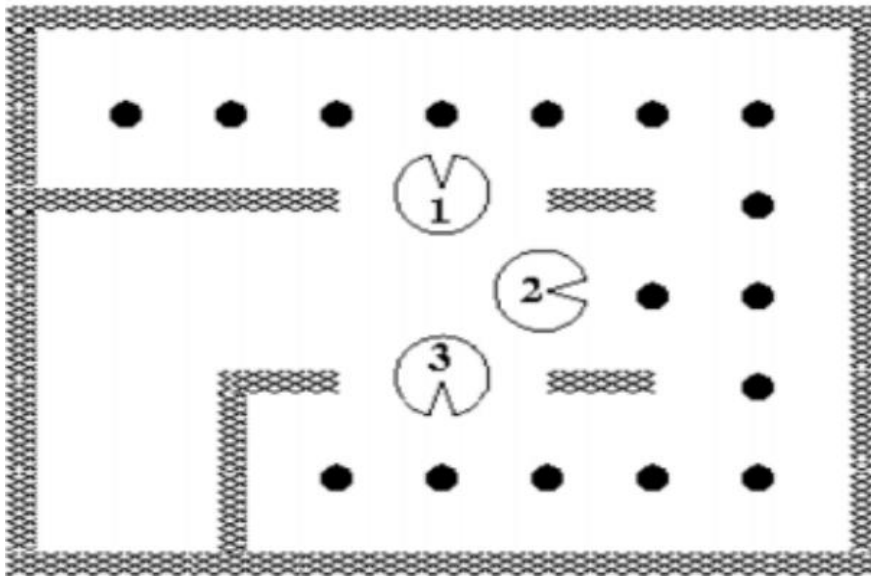
Legal actions (given a state):

Successor function (given a state and an action):

- b. Give a non-trivial admissible heuristic for this problem.
- c. Circle all of the following graph search methods which are guaranteed to output optimal solutions to this problem:
- (i) DFS
 - (ii) BFS
 - (iii) UCS
 - (iv) A* (with a consistent and admissible heuristic)
 - (v) A* (with heuristic that returns zero for each state)
- d. If h_1 and h_2 are admissible, which of the following are also guaranteed to be admissible? Circle all that apply:
- (i) $h_1 + h_2$
 - (ii) $h_1 * h_2$
 - (iii) $\max(h_1, h_2)$
 - (iv) $\min(h_1, h_2)$
 - (v) $(\alpha)h_1 + (1-\alpha)h_2$ for any value α between 0 and 1

4) (Multifactor search)

Pacman is trying eat all the dots, but he now has the help of his family! There are initially k dots, at positions $(f_1, \dots, f_k)$. There are also n Pac-People, at positions $(p_1, \dots, p_n)$; initially, all the Pac-People start in the bottom left corner of the maze. Consider a search problem in which all Pac-People move simultaneously; that is, in each step each Pac-Person moves into some adjacent position (N, S, E, or W, no STOP). Note that any number of Pac-People may occupy the same position.



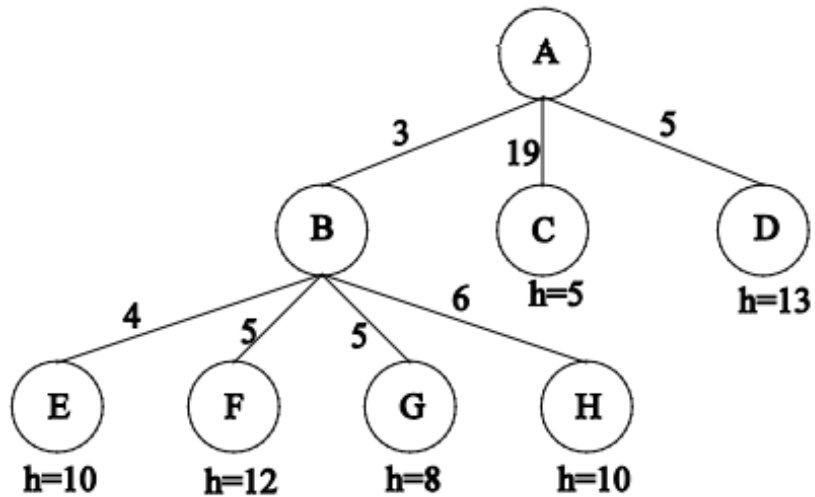
- Define the state space of the search problem.
- Give a reasonable upper bound on the size of the state space for a general r by c grid.
- What is the goal test?
- What is the maximum branching factor of the successor function in a general grid?

(e) Circle the admissible heuristics below (−1/2 point for each mistake.)

- $h_1(s) = 0$
- $h_2(s) = 1$
- $h_3(s) = \text{number of remaining food} / n$
- $h_4(s) = \max_i \max_j \text{manhattan}(p_i, f_j)$
- $h_5(s) = \max_i \min_j \text{manhattan}(p_i, f_j)$

5) (Graph search algorithms)

Consider the following search tree produced after expanding nodes A and B, where each arc is labeled with the cost of the corresponding operator, and the leaves are labeled with the value of a heuristic function, h . For uninformed searches, assume children are expanded left to right. In case of ties, expand in alphabetical order.



Which one node will be expanded next by each of the following search methods?

- (a) Depth-First search
- (b) Greedy Best-First search
- (c) Uniform-Cost search
- (d) A* search

6) (search algorithms)

True or False:

- (a) Greedy Best-First search using an admissible heuristic is guaranteed to find an optimal solution.
- (b) Algorithm A search using the heuristic $h(n) = c$ for some fixed constant $c > 0$ is guaranteed to find an optimal solution.
- (c) If a heuristic is consistent, it is also admissible.
- (d) If h_1 and h_2 are both admissible heuristics, it is always better to use the heuristic $h_3(n) = \max(h_1(n), h_2(n))$ rather than the heuristic $h_4(n) = \min(h_1(n), h_2(n))$.
- (e) Beam search with a beam width $W = 3$ and an admissible heuristic is not guaranteed to find a solution (optimal or not) when one exists.
- (f) Say we have a state space where all arc costs are 1 but we don't have a heuristic function. We want to use a space-efficient search algorithm (in terms of the maximum number of nodes stored at any point during the search in Frontier) but also want to guarantee that we find an optimal solution. Which one of the following search methods would be best to use in this situation?
 - (i) Breadth-First Search
 - (ii) Depth-First Search
 - (iii) Iterative-Deepening Search
 - (iv) Uniform-Cost Search